Modern Physics: Study Notes

Unit 3: Waves and Ultrasonic — Diploma in Engineering Course

Topic 1: Wave Motion and Types of Waves

- **Definition: Wave motion** is a periodic disturbance that travels through a medium, transferring energy from one point to another without transferring any actual matter.
- **3 Features:**
  - Waves are classified into two main types: **longitudinal** and **transverse** waves.
  - In transverse waves, particles vibrate up and down, perpendicular to the direction the wave travels.
  - In longitudinal waves, particles vibrate back and forth, parallel to the direction the wave travels.
- **3 Uses:**
  - Explains how sound energy travels through fluids, solids, and air columns.
  - Helps engineers design structural buildings that can absorb destructive seismic waves.
  - Used to transmit telecom signals over massive distances via air or glass cables.
- **Real-Life Application:** When you drop a stone in a calm pond, ripples move outward across the water surface as a transverse wave, carrying energy away from the splash point.

Topic 2: Wave Parameters and Relationships

- **Definition:** Wave parameters are measurable characteristics that describe a wave's shape and speed, including **frequency, periodic time, amplitude, wavelength, and wave velocity**.
- **3 Features:**
  - Wavelength is the physical distance between two consecutive peaks or crests of a wave.
  - Frequency is the number of complete wave cycles that pass a point in one single second.
  - They are tied together by the universal formula: **Wave Velocity = Frequency × Wavelength**.
- **3 Uses:**
  - Helps tuning technicians adjust musical instruments to correct sound frequencies.
  - Used by radio engineers to calibrate specific broadcasting channels and bandwidths.
  - Allows acoustic engineers to calculate how fast sound travels through different materials.
- **Real-Life Application:** Adjusting your car radio dial changes the target reception frequency so it perfectly matches the incoming broadcast wavelength.

Topic 3: Ultrasonic Sound and Production Methods

- **Definition: Ultrasonic sound** is a high-frequency sound wave that is above 20,000 Hertz, making it completely silent and inaudible to human ears.
- **3 Features:**
  - Carries immense energy and has a very short wavelength compared to audible sounds.
  - Can travel long distances through dense mediums without losing much of its path direction.
  - Produced in labs using specific techniques like the **Piezoelectric method** or **Magnetostriction method**.
- **3 Uses:**
  - Used to mix industrial liquids smoothly that normally don't combine well.
  - Helps clean delicate, tiny mechanism parts like watch gears or medical jewelry.
  - Used to drill precise, micro-sized holes into hard blocks of glass or ceramics.
- **Real-Life Application:** Medical doctors use ultrasound machines to see clear live pictures of an unborn baby growing safely inside a mother's womb.

Topic 4: SONAR and Non-Destructive Testing (NDT)

- **Definition: SONAR** stands for Sound Navigation and Ranging, and **NDT** is a safety method used to inspect industrial materials for deep internal cracks without damaging them.
- **3 Features:**
  - Works on the basic principle of echoing where ultrasonic waves bounce off hidden surfaces.
  - The time a sound takes to return reveals the exact depth or location of a hidden object.
  - Completely safe and does not alter or weaken the physical material being inspected.
- **3 Uses:**
  - Allows ships and submarines to navigate deep waters safely and avoid crashing.
  - Scans underwater seafloors to draw highly accurate topographical ocean maps.
  - Inspects steel railway tracks and airplane bodies to detect microscopic hidden cracks.
- **Real-Life Application:** Submarines send out silent ultrasound pings into the dark ocean water to detect how close they are to underwater mountains or nearby vessels.